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Ans to the Q No. 1

i) Let, L_1 be regular, P be Pumping length for L_1 ,

$w = 0^P 1^{P+10P} \in L$, Then, $|w| = 12P > P$

w is split into xyz such that, i) $|xy| \leq P$ (ii) $|y| > 0$ (iii) $xy^iz \in L \forall i \geq 0$

Let,

$$x = 0^\alpha$$

$$y = 0^\beta$$

$$z = 0^{P-\alpha-\beta} 1^{P+10P}$$

Now,

$$xy^iz = 0^\alpha 0^{2\beta} 0^{P-\alpha-\beta} 1^{P+10P}$$
$$= 0^{P+\beta} 1^{P+10P}$$

Now, if $xy^iz \in L$,

then,

$$P+\beta = P$$

$$\text{or, } \beta = 0$$

But $|y| > 0$ so, $xy^iz \notin L$,

$\therefore L_1$ is not a regular Language

ii) Let,

$$0^P 110^P \in L$$

$$x = 0^{P-\alpha}$$

$$y = 0^\alpha$$

$$z = 110^P$$

Now,

$$xy^iz = 0^{P-\alpha} 0^{2\alpha} 110^P$$
$$= 0^{P+\alpha} 110^P$$

if $xy^iz \in L_2$ then,

$$P+\alpha = P$$

$$\text{or, } \alpha = 0$$

But, $|y| > 0$, so $xy^iz \notin L_2$

$\therefore L_2$ is not a regular Language

iii) Let, L_3 be regular Language

So, $\bar{L}_3$ will also be regular Language

Let, w be $0^P 10^P$

$$x = 0^{P-\alpha}$$

$$y = 0^\alpha$$

$$z = 10^P$$

if $xy^iz \in \bar{L}_3$

$$\text{then, } 0^{P-\alpha} 0^{2\alpha} 10^P$$
$$= 0^{P+\alpha} 10^P$$

if $xy^iz \in \bar{L}_3$

then, $P+\alpha = P$

$$\text{or, } \alpha = 0$$

But, $|y| > 0$ so $xy^iz \notin \bar{L}_3$

$\therefore \bar{L}_3$ is not regular Language

So, L_3 will also not be regular Language

Ans to the Q NO. 2

$$D) S \rightarrow OS3 | OA3$$

$$A \rightarrow 1A2 | 12$$

$$1) S \xrightarrow{ABC} \cancel{ABC} | OBC | AB2$$

$$A \rightarrow OA | \epsilon$$

$$C \rightarrow C2 | \epsilon$$

$$B \rightarrow O1B | B12 | \epsilon$$

$$11) S \rightarrow \epsilon$$

$$S \rightarrow A00S11A | 1S | \cancel{S}SO | X\#X$$

$$A \rightarrow 01 | \epsilon$$

$$X \rightarrow 0 | 1 | \epsilon$$

Ans to the Q NO. 3

$$q_1 \rightarrow 0q_3 | 1q_2 | \epsilon$$

$$q_2 \rightarrow 0q_5 | 1q_5 | \epsilon$$

$$q_3 \rightarrow 1q_1 | 0q_4$$

$$q_4 \rightarrow 1q_4 | 0q_2$$

$$q_5 \rightarrow 0q_5 | 1q_5$$

Ans to the Q No. 4

$$a) S = abS$$

$$= abbaS$$

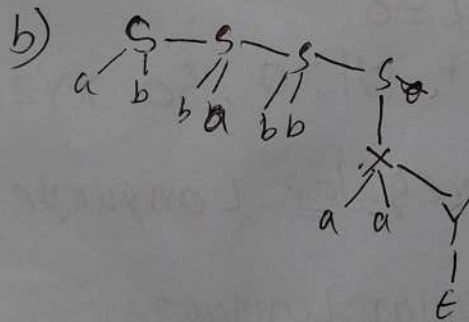
$$= abbabbs$$

$$= abbabbbx$$

$$= abbabbaay$$

$$= abbabbaae$$

$$= abbiabbaa$$



$$c) S \Rightarrow abS \Rightarrow abx$$

$$\Rightarrow abbaY$$

$$\Rightarrow abbabby$$

$$\Rightarrow abbabbaY$$

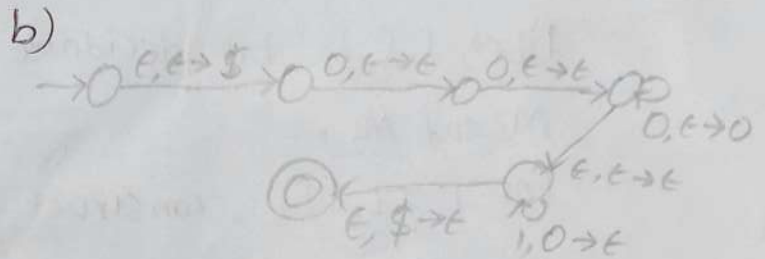
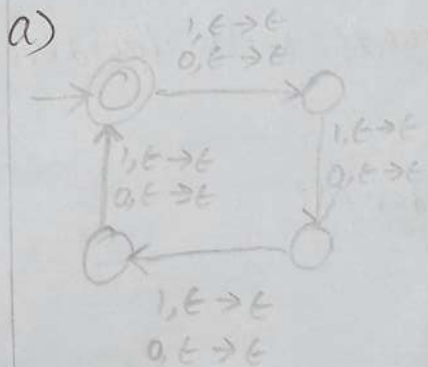
$$\Rightarrow abbabbaay$$

$$\Rightarrow abbabbaa$$

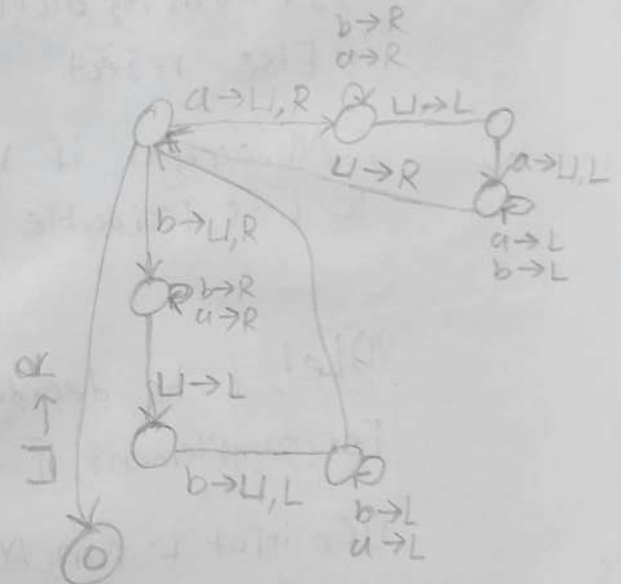
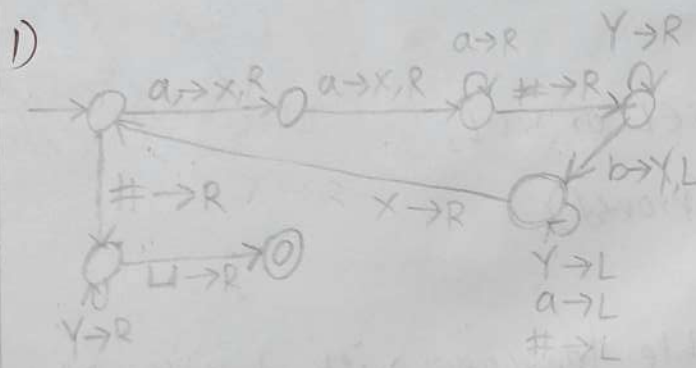
$\therefore$ the grammar is ambiguous

$$d) bbbbaa$$

Ans to the Q NO. 5



Ans to the Q NO. 6



Ans to the Q NO. 7

1) $M_H =$ "On input $\langle M, w \rangle$:

- 1) Simulate M on w for at most N steps.
- 2) If M halts with in N step, accept else reject.

Since simulation runs on finite N steps, so M_H is decider (Proved)

1) $M_G =$ "On input $\langle G \rangle$:

- 1) Perform BFS or DFS from a vertex, mark it visited
 - 2) If all vertex are visited accept else rejected
- Since G is finite, M_G is decider (Proved)

Ans to the Q NO. 8

1) Let, L_1 & L_2 be decidable language with deciders M_1 and M_2 .

For $L = L_1 \cup L_2$, construct decider M :

- 1) On input w , run M_1 on w
- 2) If M_1 reject run M_2 on w
- 3) If M_1 or M_2 accept, M accept
- 4) Else reject

$\therefore M$ accepts if $w \in L_1$ or $w \in L_2$

So L is decidable (proved)

11) Let, L be decidable language with decider M
For complement, $\bar{L} = \Sigma^* \setminus L$, construct decider M'

- 1) On input w run M on w
- 2) If M accept then reject w
- 3) If M reject then accept w

M' always halts and accepts if $w \notin L$
So $\bar{L}$ is decidable. (proved)